MIT | Academy of Engineering

(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

ASSIGNMENT 1

Analysis of Artificial Intelligence Agents, Environments, and Their Ethical & Societal Implications in Real-World Applications

MIT Academy of Engineering

(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Alandi (D), Pune – 412105 (2026–

2027)

"To study and analyze the application of Artificial Intelligence and Machine Learning in real-world systems by examining different types of intelligent agents and environments, and to evaluate their PEAS characteristics, societal impact, ethical and legal concerns, privacy, surveillance, bias, discrimination, and the role of human judgment."

Name – Rushikesh Ganesh Shinde.

PRN – 202401040040

Div – A , Batch – A1

Roll No – 9

Course – AIML

1. Artificial Intelligence in Autonomous Ride-Hailing

1. Basic Information About the Application

1.1 Introduction

Autonomous ride-hailing is a modern application of Artificial Intelligence (AI) in which a vehicle can transport passengers with little or no human driving.
A passenger can request a ride using a mobile application. The autonomous vehicle then travels to the pickup location, picks up the passenger and drives to the destination.
One real-world example is **Waymo's autonomous driving technology**. The Waymo Driver uses AI, sensors, maps and computing systems to understand the road and make driving decisions.
The vehicle needs to perform many intelligent tasks such as:

- Detecting vehicles
- Detecting pedestrians
- Recognizing traffic signals
- Understanding road conditions
- Selecting a route
- Maintaining safe speed
- Avoiding obstacles
- Predicting movement of other road users
- Making driving decisions
- Learning from previous experience

Therefore, autonomous ride-hailing is a good real-world example for studying **AI agents, environments, PEAS, Machine Learning, ethical issues and societal impact**.

1.2 How AI Works in Autonomous Ride-Hailing

The basic working process is:
Sensors → Data Processing → AI Decision → Vehicle Action
For example:
Camera/Lidar detects pedestrian → AI identifies pedestrian → AI calculates danger → Vehicle slows/stops
The AI system continuously observes the environment and takes appropriate actions.

---

2. AI Agent Types Used in This Report

Agent Type	Simple Meaning
Simple Reflex Agent	Acts according to the current situation
State-Based Agent	Uses current and previous information
Goal-Based Agent	Takes actions to achieve a goal
Utility-Based Agent	Selects the best action among alternatives
Learning Agent	Learns and improves from experience

---

3. Agent–Environment Analysis

The following 15 sections use different combinations of the five agents and five environments.

1) **Safe City Driving Using a Simple Reflex Agent**

Agent
Simple Reflex Agent
Environment
Urban Road
Application
The autonomous vehicle is travelling through a city road.
The vehicle continuously checks the current road condition and immediately reacts to it.
Example

- If the traffic light is red → Stop.
- If a pedestrian is detected → Slow down/stop.
- If an obstacle is detected → Brake.
- If the road is clear → Continue.

PEAS

PEAS	Description
Performance	Safety, traffic-rule following and collision avoidance
Environment	City roads
Actuators	Brake, accelerator and steering
Sensors	Camera, lidar, radar and GPS

Environment Characteristics

Property	Type
Observable	Partially observable
Deterministic	Non-deterministic
Static	Dynamic
Agents	Multi-agent
Continuous	Yes
Known	Partially known

Positive Societal Impact

- Can improve road safety.
- Can reduce human driving mistakes.
- Can provide convenient transportation.
- Can help people who cannot drive.

Negative Societal Impact

- May reduce employment opportunities for drivers.
- People may become dependent on autonomous vehicles.
- AI mistakes can cause serious accidents.

Ethical Issues

- Who is responsible if the vehicle makes a wrong decision?
- How should passenger safety and pedestrian safety be balanced?
- How should vehicle data be protected?

---

2) Emergency Highway Response

Agent
Simple Reflex Agent
Environment
Highway
Application
The vehicle is travelling at high speed and suddenly detects an obstacle.
The agent must react immediately.
Example
Obstacle detected → Brake
Vehicle too close → Slow down
Dangerous situation → Take immediate safe action

PEAS

PEAS	Description
Performance	Fast and safe reaction
Environment	Highway
Actuators	Brakes, steering, accelerator
Sensors	Camera, lidar and radar

Environment Characteristics

- Dynamic
- Partially observable
- Continuous
- Non-deterministic
- Multi-agent

Positive Impact

- Fast response to sudden obstacles.
- Can reduce accidents caused by delayed human reaction.

Negative Impact

- False detection may cause unnecessary braking.
- Sudden braking may affect vehicles behind it.

Ethical Issue

The system should always give priority to **human safety rather than speed or passenger convenience**.

---

3) Intelligent Traffic-Signal Response

Agent
Simple Reflex Agent
Environment
Intersection
Application
The vehicle approaches a traffic signal.
Rules

- Red signal → Stop.
- Green signal → Move if road is clear.
- Pedestrian detected → Stop.
- Collision risk → Brake.

PEAS

PEAS	Description
Performance	Safe intersection crossing
Environment	Traffic intersection
Actuators	Brake, accelerator, steering
Sensors	Cameras, lidar, radar

Environment Characteristics

- Dynamic
- Partially observable
- Multi-agent
- Continuous
- Uncertain

Positive Impact

- Better traffic-signal compliance.
- Safer pedestrian crossings.

Negative Impact

- Sensor failure may cause wrong decisions.
- Incorrect signal detection can cause accidents.

Ethical Issue

The AI should not blindly trust one sensor. Important decisions should be supported by reliable sensor information.

---

4) Understanding City Traffic with a State-Based Agent

Agent
State-Based Agent
Environment
Urban Road
Application

A state-based agent uses current information as well as previous information.
For example:
A car is detected beside the autonomous vehicle.
The car temporarily disappears behind another vehicle.
The system can remember:
"There was a vehicle in that position."

PEAS

PEAS	Description
Performance	Correct understanding of surrounding traffic
Environment	Urban road
Actuators	Steering, braking, acceleration
Sensors	Camera, lidar, radar, GPS

Environment Characteristics

- Partially observable
- Dynamic
- Non-deterministic
- Multi-agent

Positive Impact

- Better understanding of surrounding vehicles.
- Better handling of temporarily hidden objects.

Negative Impact

If the internal state is incorrect, the vehicle may make a wrong decision.

Ethical Issue

The AI should recognize uncertainty rather than treating its prediction as 100% correct.

---

5) Smart Parking Using State-Based Agent

Agent
State-Based Agent
Environment
Parking Area
Application

The vehicle enters a parking area.
It must remember:

- Its current position
- Available parking spaces
- Nearby vehicles
- Walls
- Pedestrians
- Previous movement

PEAS

PEAS	Description
Performance	Safe and accurate parking
Environment	Parking area
Actuators	Steering, brake, accelerator
Sensors	Camera, lidar, radar

Environment Characteristics

- Partially observable
- Dynamic
- Continuous
- Multi-agent

Positive Impact

- Easier parking.
- Reduces parking mistakes.
- Helps elderly and disabled passengers.

Negative Impact

- Sensor errors may cause collisions.
- Excessive dependence on automated parking.

Ethical Issue

Pedestrian safety should be more important than quickly finding a parking space.

---

6) Driving in Rain and Fog

Agent
State-Based Agent
Environment
Adverse Weather
Application

The vehicle operates during:

- Heavy rain
- Fog
- Poor visibility

The agent maintains information about the environment and uses available sensor information to make decisions.

PEAS

PEAS	Description
Performance	Safe driving in poor visibility
Environment	Rain/fog
Actuators	Brakes, steering, accelerator
Sensors	Camera, lidar, radar

Environment Characteristics

- Partially observable
- Dynamic
- Uncertain
- Continuous

Positive Impact

- Can reduce dangerous driving during bad weather.
- Can maintain continuous monitoring.

Negative Impact

- Sensors may become less reliable.
- Poor visibility can increase risk.

Ethical Issue

The vehicle should slow down or stop safely if it cannot reliably understand the environment.

Safety guidance for automated driving systems emphasizes areas including object detection, fallback/minimal-risk conditions and cybersecurity.

---

7) Reaching the Destination Safely
Agent
Goal-Based Agent
Environment
Highway
Goal
Reach the passenger's destination safely.
The vehicle chooses actions that help achieve the destination.

PEAS

PEAS	Description
Performance	Safe arrival at destination
Environment	Highway
Actuators	Steering, braking, acceleration
Sensors	GPS, cameras, lidar, radar

Environment Characteristics
- Dynamic
- Continuous
- Multi-agent
- Partially observable
- Non-deterministic

Positive Impact
- Convenient transportation.
- Reduced driving workload.
- Better mobility for passengers.

Negative Impact
- Possible job losses for professional drivers.
- Technical failure can affect passenger safety.

Ethical Issue
The goal of reaching the destination should **never be more important than human safety**.

5.8 Safe Movement Through a Busy Intersection
Agent
Goal-Based Agent
Environment
Intersection
Goal
Cross the intersection safely and continue toward the destination.

Possible Actions

- Stop
- Wait
- Slow down
- Continue
- Change lane

PEAS

PEAS	Description
Performance	Safe movement and destination progress
Environment	Busy intersection
Actuators	Steering, brakes, accelerator
Sensors	Cameras, lidar, radar

Environment Characteristics

- Dynamic
- Multi-agent
- Partially observable
- Non-deterministic

Positive Impact

- Better intersection management.
- Increased pedestrian safety.

Negative Impact

- Incorrect prediction of other vehicles can create danger.

Ethical Issue

Even if another driver has broken a traffic rule, the autonomous vehicle should try to avoid a collision.

---

9) Reaching Destination in a Pedestrian Area

Agent

Goal-Based Agent

Environment

Pedestrian Environment

Goal

Reach the passenger's destination while protecting pedestrians.

PEAS

PEAS	Description
Performance	Safe destination arrival
Environment	Pedestrian-heavy road
Actuators	Steering, brake, accelerator
Sensors	Camera, lidar, radar

Environment Characteristics

- Highly dynamic
- Partially observable
- Multi-agent
- Uncertain

Positive Impact

- Increased pedestrian protection.
- Convenient transportation.

Negative Impact

- Vehicle may need to move slowly.
- Traffic may increase if vehicles are overly cautious.

Ethical Issue

This environment creates an important question:
If reaching the destination takes longer but provides greater safety, should the AI choose the safer option?
Yes. Human safety should have priority.

---

10) Balancing Safety and Travel Time

Agent
Utility-Based Agent
Environment
Urban Road
A utility-based agent compares different choices.
For example:

Route	Time	Safety	Comfort
Route A	Short	Low	Medium
Route B	Medium	High	High
Route C	Long	Very High	Medium

The system should select the option that provides the best overall result, while treating safety as a critical requirement.
PEAS

PEAS	Description
Performance	Safety, time and comfort
Environment	Urban road
Actuators	Steering, brakes, accelerator
Sensors	GPS, cameras, lidar, radar

Environment Characteristics

- Dynamic
- Continuous
- Multi-agent
- Uncertain

Positive Impact

- Better route selection.
- Reduced unnecessary travel.
- Improved passenger experience.

Negative Impact

- Incorrect utility calculation could produce unsafe decisions.

Ethical Issue

Saving a few minutes should not justify significantly increasing safety risk.

---

11) Efficient Highway Travel

Agent

Utility-Based Agent

Environment

Highway

The agent considers:

- Safety
- Travel time
- Traffic
- Passenger comfort
- Energy consumption

PEAS

PEAS	Description
Performance	Safe and efficient travel
Environment	Highway
Actuators	Steering, brakes, accelerator
Sensors	GPS, camera, lidar, radar

Environment Characteristics

- Dynamic
- Continuous
- Multi-agent
- Non-deterministic

Positive Impact

- Efficient travel.
- Better route selection.
- Potential energy savings.

Negative Impact

- Optimization for speed may create safety risks.

Ethical Issue

The system must prioritize safety over speed.

---

12) Safe Driving During Bad Weather

Agent

Utility-Based Agent

Environment

Adverse Weather

The agent compares:

- Continue normally
- Reduce speed
- Change route
- Stop temporarily

PEAS

PEAS	Description
Performance	Safety and travel efficiency
Environment	Rain/fog
Actuators	Brake, steering, accelerator
Sensors	Camera, lidar, radar

Environment Characteristics

- Partially observable
- Dynamic
- Uncertain
- Continuous

Positive Impact

- Safer driving decisions.
- Better response to weather changes.

Negative Impact

- Journey may become longer.
- Service availability may decrease during severe weather.

Ethical Issue

Passenger convenience should not be prioritized over public safety.

---

13) Learning Better Parking

Agent

Learning Agent

Environment

Parking Area
The agent learns from previous parking experiences.
It can learn:

- How much to turn the steering wheel.
- How close the vehicle is to obstacles.
- How to enter a parking space.
- How to leave the parking space safely.

PEAS

PEAS	Description
Performance	Accurate and safe parking
Environment	Parking area
Actuators	Steering, brake, accelerator
Sensors	Camera, lidar, radar

Environment Characteristics

- Dynamic
- Partially observable
- Continuous
- Multi-agent

Positive Impact

- Improved parking performance.
- Fewer parking errors.

Negative Impact

- Poor training data can produce poor performance.
- Learning from unusual situations may be difficult.

Ethical Issue
The training data should represent different parking environments and conditions.

---

14) Learning Pedestrian Behaviour
Agent
Learning Agent
Environment
Pedestrian Environment

The AI can learn patterns of pedestrian movement.
For example:

- People waiting before crossing
- People crossing at zebra crossings
- Groups of people crossing together
- Unexpected pedestrian movement

PEAS

PEAS	Description
Performance	Accurate pedestrian prediction
Environment	Pedestrian-heavy area
Actuators	Steering, brake, accelerator
Sensors	Cameras, lidar, radar

Environment Characteristics

- Highly dynamic
- Uncertain
- Partially observable
- Multi-agent

Positive Impact

- Better pedestrian safety.
- Better prediction of road behaviour.

Negative Impact

If training data is incomplete, the system may perform differently for different people or situations.

Ethical Issue

This can create **bias and discrimination**.

For example, the system should be tested with:

- Children
- Adults
- Elderly people
- Wheelchair users
- Cyclists
- People in different clothing
- Different lighting conditions

---

15) Learning Better Decisions at Intersections

Agent
Learning Agent
Environment
Intersection

The AI can learn from previous driving situations.

It can learn:

- Traffic patterns
- Pedestrian movement
- Vehicle movement
- Common traffic situations
- Difficult intersections

PEAS

PEAS	Description
Performance	Safe and accurate decision-making
Environment	Complex intersection
Actuators	Steering, braking, acceleration
Sensors	Camera, lidar, radar, GPS

Environment Characteristics

- Dynamic
- Multi-agent
- Partially observable
- Non-deterministic
- Continuous

Positive Impact

- Better decision-making.
- Improved traffic handling.
- Continuous improvement.

Negative Impact

- The AI may learn incorrect patterns.
- Rare situations may not be represented sufficiently in training data.

Ethical Issue

The AI should not assume that because something usually happens, it will always happen.
Rare situations can be extremely important for safety.

---

4. Common Societal Impact of Autonomous Ride-Hailing

Some effects apply to the complete application rather than only one agent.

Positive Societal Impact

1. Improved Safety

AI can continuously monitor the road and react to hazards.

2. Better Accessibility

Autonomous transportation may help people who cannot drive.

3. Convenience

Passengers can travel without manually driving.

4. Reduced Human Error

AI does not become tired or distracted in the same way as human drivers.

5. Better Transportation

AI can help select routes and manage transportation efficiently.

---

5. Negative Societal Impact

1. Employment
Autonomous vehicles could affect jobs such as:

- Taxi drivers
- Cab drivers
- Delivery drivers
- Other professional drivers

2. Privacy
Vehicles may collect information about:

- Location
- Travel routes
- Pickup points
- Drop-off points

NHTSA identifies privacy, including precise real-time geolocation, as an important consideration for automated driving systems.
3. Surveillance
Vehicle sensors can observe people, vehicles and public areas.
4. Cybersecurity
A cyberattack against an autonomous vehicle could become a safety problem.
5. AI Bias
Incorrect or unbalanced training data can result in different performance in different situations.
6. Technology Dependence
People may become highly dependent on autonomous transportation.

---

6. Ethical Concerns

6.1 Privacy
Passenger travel information should be protected.
6.2 Surveillance
Continuous sensing of public areas should be controlled.
6.3 Bias
AI should be tested with diverse data.
6.4 Discrimination
AI should not provide lower-quality performance for particular groups.
6.5 Safety
Safety must be the highest priority.
6.6 Accountability
There must be clear responsibility when an autonomous system fails.
6.7 Transparency
Users should know the basic capabilities and limitations of the system.
6.8 Cybersecurity
The vehicle must be protected from unauthorized access.

7. Final Conclusion

Artificial Intelligence is playing an important role in the development of autonomous ride-hailing systems.
This report studied the application using **five AI agents and five different environments**, resulting in **15 different agent–environment cases**.
The analysis shows that different AI agents are suitable for different situations. Simple reflex agents can handle immediate reactions, state-based agents can remember previous information, goal-based agents can work toward destinations, utility-based agents can balance multiple factors, and learning agents can improve using experience.
Autonomous ride-hailing can provide several benefits such as:

- Improved transportation
- Better accessibility
- Convenience
- Potential safety improvements
- Reduced human driving workload

However, it also creates important concerns such as:

- Privacy
- Surveillance
- Cybersecurity
- Employment
- Bias
- Discrimination
- Accountability
- Human judgment

The main goal of autonomous AI should not simply be to make vehicles drive without humans, but to make transportation safer, more reliable, ethical and beneficial for society.

2. Artificial Intelligence in AI-Based Medical Diagnosis

1. Basic Information About the Application

1.1 Introduction

AI-Based Medical Diagnosis is an important application of Artificial Intelligence and Machine Learning in healthcare.
In this application, AI systems analyze medical information such as:

- Medical images
- Patient symptoms
- Blood test results
- Medical history
- ECG data
- X-ray images
- CT scans
- MRI scans
- Laboratory reports

The AI system helps doctors identify possible diseases and make better decisions.
For example:
Medical Image → AI analyzes image → Disease pattern detected → Possible diagnosis → Doctor reviews result
AI can be used for detecting diseases such as:

- Cancer
- Pneumonia
- Diabetes-related conditions
- Heart diseases
- Eye diseases
- Skin diseases

The purpose of AI in medical diagnosis is generally to **assist healthcare professionals**, not simply replace them.

1.2 How AI Works in Medical Diagnosis

The basic process is:
Patient Data → Data Processing → AI Analysis → Prediction → Doctor's Decision

2. AI Agent Types Used in This Report

Agent Type	Simple Meaning
Simple Reflex Agent	Acts according to the current medical information
State-Based Agent	Uses current and previous patient information
Goal-Based Agent	Takes actions to achieve a diagnosis-related goal
Utility-Based Agent	Selects the most useful diagnostic option
Learning Agent	Learns and improves from medical data

---

3. Agent–Environment Analysis

1) Quick Disease Detection in Hospital Using a Simple Reflex Agent

Agent
Simple Reflex Agent
Environment
Hospital
Application
The AI system checks the current patient information and immediately responds according to predefined conditions.
Example

- If temperature is very high → Alert medical staff.
- If oxygen level is dangerously low → Generate emergency alert.
- If blood pressure is extremely abnormal → Notify doctor.
- If a specific symptom pattern is detected → Suggest further examination.

PEAS

PEAS	Description
Performance	Fast and accurate detection of abnormal conditions
Environment	Hospital
Actuators	Alerts, notifications and diagnostic suggestions
Sensors	Patient records, vital-sign monitors and medical inputs

Environment Characteristics

Property	Type
Observable	Partially observable
Deterministic	Non-deterministic
Static	Dynamic
Agents	Multi-agent
Continuous	Yes
Known	Partially known

Positive Societal Impact

- Faster identification of dangerous conditions.
- Helps doctors monitor many patients.
- Can reduce delays in emergency situations.
- Supports healthcare workers.

Negative Societal Impact

- Incorrect alerts may create unnecessary anxiety.
- Doctors may become too dependent on AI.
- AI errors can affect patient safety.

Ethical Issues

- Who is responsible if the AI gives an incorrect alert?
- Should the doctor always verify the AI result?
- How should patient information be protected?

---

2) Fast Emergency Alert Using a Simple Reflex Agent
Agent
Simple Reflex Agent
Environment
Emergency Department
Application
The AI system monitors current patient conditions and immediately produces an alert when a dangerous condition is detected.
Example
Very low oxygen level → Emergency alert
Abnormal heart rate → Notify doctor
Critical blood pressure → Immediate warning
PEAS

PEAS	Description
Performance	Fast and correct emergency detection
Environment	Emergency department
Actuators	Alerts and notifications
Sensors	ECG, blood pressure, oxygen and patient-monitoring data

Environment Characteristics

- Dynamic
- Partially observable
- Continuous
- Multi-agent
- Non-deterministic

Positive Impact

- Faster emergency response.
- Helps doctors identify high-risk patients.
- Can continuously monitor patients.

Negative Impact

- False alarms may waste medical resources.
- Incorrect detection may delay proper treatment.

Ethical Issue

The AI should assist doctors but should not make unsafe medical decisions without appropriate clinical verification.

---

3) Immediate Medical Test Analysis

Agent

Simple Reflex Agent

Environment

Diagnostic Laboratory

Application

The system checks laboratory results against predefined medical ranges.

Example

- If glucose is above a specified range → Flag abnormal result.
- If hemoglobin is below a specified range → Flag result.
- If a test is highly abnormal → Notify medical staff.

PEAS

PEAS	Description
Performance	Correct identification of abnormal results
Environment	Diagnostic laboratory
Actuators	Alerts and report markings
Sensors	Blood-test and laboratory data

Environment Characteristics

- Partially observable
- Dynamic
- Continuous
- Multi-agent
- Partially known

Positive Impact

- Faster identification of abnormal results.
- Reduces manual checking.
- Helps laboratory staff.

Negative Impact

- A reference range alone may not provide complete clinical context.
- False alerts may occur.

Ethical Issue

An abnormal test result should not automatically be treated as a final diagnosis.

---

4) Understanding Patient History Using a State-Based Agent
Agent
State-Based Agent
Environment
Hospital
Application
The system considers both current and previous patient information.
For example, the patient may have:

- Previous diseases
- Previous test results
- Previous medications
- Previous symptoms
- Current symptoms

The AI maintains an internal state of the patient's medical history.
PEAS

PEAS	Description
Performance	Correct understanding of patient history
Environment	Hospital
Actuators	Alerts, reports and recommendations
Sensors	Electronic health records, test reports and patient information

Environment Characteristics

- Partially observable
- Dynamic
- Non-deterministic
- Multi-agent

Positive Impact

- Provides doctors with a better patient history.
- Reduces the need to manually check multiple records.
- Can identify changes in patient condition.

Negative Impact

- Incorrect or incomplete records may affect the AI's decision.
- Old information may be incorrectly considered relevant.

Ethical Issue
Patient information must be accurate, private and properly controlled.

---

5) Tracking Patient Health in the Hospital
Agent
State-Based Agent
Environment
Hospital
Application
The system tracks the patient's changing condition over time.

For example:
Morning: Normal temperature
Afternoon: Increased temperature
Evening: High temperature
The AI can use this history to identify a change in condition.

PEAS

PEAS	Description
Performance	Correct monitoring of patient condition
Environment	Hospital
Actuators	Alerts and reports
Sensors	Patient monitors and medical records

Environment Characteristics

- Dynamic
- Partially observable
- Continuous
- Multi-agent

Positive Impact

- Continuous patient monitoring.
- Early detection of changes.
- Supports doctors and nurses.

Negative Impact

- Incorrect sensor data can produce incorrect conclusions.
- Excessive monitoring can create privacy concerns.

Ethical Issue
Continuous patient monitoring should be limited to legitimate healthcare purposes.

---

6) Monitoring Patient Results Over Time
Agent
State-Based Agent
Environment
Diagnostic Laboratory
Application
The AI compares the patient's current laboratory results with previous results.
For example:
Previous blood test → Current blood test → Compare changes → Alert doctor if significant
PEAS

PEAS	Description
Performance	Correct identification of changes
Environment	Diagnostic laboratory
Actuators	Alerts and reports
Sensors	Laboratory results and patient history

Environment Characteristics

- Partially observable
- Dynamic
- Continuous
- Multi-agent

Positive Impact

- Helps identify trends.
- Reduces manual comparison.
- Helps doctors understand changes in patient condition.

Negative Impact

- Incorrect historical information can produce wrong conclusions.
- Correlation does not always mean disease.

Ethical Issue

AI predictions should be reviewed by qualified medical professionals.

---

7) Finding a Possible Diagnosis

Agent

Goal-Based Agent

Environment

Hospital

Goal

Identify the most likely medical condition using available patient information.

The AI considers:

- Symptoms
- Medical history
- Test results
- Age
- Other relevant information

PEAS

PEAS	Description
Performance	Accurate and useful diagnostic support
Environment	Hospital
Actuators	Diagnostic suggestions and reports
Sensors	Patient records, symptoms and test results

Environment Characteristics

- Partially observable
- Dynamic
- Non-deterministic
- Multi-agent
- Continuous

Positive Impact

- Helps doctors identify possible diseases.
- Reduces time spent searching through large amounts of information.
- Supports clinical decision-making.

Negative Impact

- AI may produce an incorrect diagnosis.
- Doctors may rely too much on AI.

Ethical Issue

The AI should support the doctor rather than independently replacing professional medical judgment.

---

8) Selecting Appropriate Diagnostic Tests

Agent

Goal-Based Agent

Environment

Diagnostic Laboratory

Goal

Identify the tests that can provide useful information about the patient's condition. For example:

Symptoms → Possible disease → Suggested test → Result → Further analysis

PEAS

PEAS	Description
Performance	Appropriate test selection
Environment	Diagnostic laboratory
Actuators	Test recommendations and reports
Sensors	Symptoms, medical history and previous test results

Environment Characteristics

- Partially observable
- Dynamic
- Multi-agent
- Non-deterministic

Positive Impact

- Can reduce unnecessary testing.
- Can help doctors select relevant tests.
- Saves time.

Negative Impact

- Incorrect recommendations can delay diagnosis.
- Unnecessary testing can increase healthcare costs.

Ethical Issue

The system should not recommend tests only because they generate more data or revenue. Medical necessity should be the priority.

---

9) Detecting Disease from Medical Images
Agent
Goal-Based Agent
Environment
Radiology Department
Goal
Detect possible abnormalities in medical images.
Examples:

- X-ray
- CT scan
- MRI

PEAS

PEAS	Description
Performance	Accurate identification of abnormalities
Environment	Radiology department
Actuators	Diagnostic alerts and reports
Sensors	X-ray, CT and MRI images

Environment Characteristics

- Partially observable
- Static during image analysis
- Continuous
- Multi-agent

Positive Impact

- Helps radiologists.
- Can identify suspicious areas quickly.
- Can reduce manual workload.

Negative Impact

- False positive results may cause unnecessary tests.
- False negative results may delay diagnosis.

Ethical Issue
AI results should be verified by qualified healthcare professionals, particularly when the consequences of an incorrect result are serious.

---

10) Choosing the Best Diagnostic Option
Agent
Utility-Based Agent
Environment
Hospital
The system compares multiple possible diagnostic options.
It can consider:

- Accuracy
- Patient risk
- Cost

- Time
- Available medical information

Example

Option	Accuracy	Time	Risk
Test A	Medium	Fast	Low
Test B	High	Medium	Medium
Test C	Very High	Slow	High

The system should select the most appropriate option according to the clinical situation.

PEAS

PEAS	Description
Performance	Safe, useful and accurate decision support
Environment	Hospital
Actuators	Recommendations and reports
Sensors	Patient information and test results

Environment Characteristics

- Dynamic
- Partially observable
- Multi-agent
- Non-deterministic

Positive Impact

- Better decision support.
- Can help reduce unnecessary procedures.
- Can save time.

Negative Impact

- Incorrect utility calculations can produce poor recommendations.
- Cost should not be allowed to override patient safety.

Ethical Issue

Patient safety must have greater importance than cost or convenience.

---

11) Balancing Accuracy and Cost

Agent

Utility-Based Agent

Environment

Diagnostic Laboratory

The AI can compare different diagnostic tests based on:

- Accuracy
- Cost
- Time
- Patient discomfort
- Medical usefulness

PEAS

PEAS	Description
Performance	Accurate and cost-effective diagnosis
Environment	Laboratory
Actuators	Test recommendations
Sensors	Medical history and laboratory data

Environment Characteristics

- Dynamic
- Partially observable
- Continuous
- Multi-agent

Positive Impact

- Can reduce unnecessary testing.
- Can improve resource utilization.
- Can reduce patient waiting time.

Negative Impact

- Choosing a cheaper test when a more accurate test is required can be harmful.
- Cost-based decisions can create inequality.

Ethical Issue

Cost should never be more important than necessary patient care.

---

12) Choosing the Best Treatment Support Option

Agent

Utility-Based Agent

Environment

Hospital

The AI can compare possible treatment-support options based on information such as:

- Expected benefit
- Patient condition
- Possible risks
- Previous medical information
- Patient requirements

PEAS

PEAS	Description
Performance	Safe and useful treatment support
Environment	Hospital
Actuators	Recommendations and alerts
Sensors	Patient records, test results and medical information

Environment Characteristics

- Partially observable
- Dynamic
- Multi-agent
- Non-deterministic

Positive Impact

- Supports doctors.
- Can help compare large amounts of medical information.
- May improve decision-making.

Negative Impact

- Incorrect recommendation can harm patients.
- Different patients may respond differently to the same treatment.

Ethical Issue

The final medical decision should remain under appropriate professional and clinical oversight.

---

13) Learning Better Disease Detection

Agent

Learning Agent

Environment

Radiology Department

The AI learns from medical images and previously confirmed cases.
It can learn patterns associated with:

- Tumors
- Infections
- Abnormal tissues
- Other medical conditions

PEAS

PEAS	Description
Performance	Accurate image-based disease detection
Environment	Radiology department
Actuators	Alerts and diagnostic support
Sensors	X-ray, CT and MRI images

Environment Characteristics

- Partially observable
- Continuous
- Multi-agent
- Data-dependent

Positive Impact

- Can improve image analysis.
- Can reduce workload for radiologists.
- Can help identify suspicious areas.

Negative Impact

- Poor-quality training data can reduce performance.
- Incorrect training labels can teach the AI wrong patterns.

Ethical Issue

Training data should be sufficiently diverse and representative.

---

14) Learning Patient Risk Patterns

Agent

Learning Agent

Environment

Emergency Department

The AI learns from previous patient data to identify patterns associated with high-risk conditions.

For example:

Symptoms + Vital Signs + Medical History → Risk Prediction

PEAS

PEAS	Description
Performance	Accurate identification of high-risk patients
Environment	Emergency department
Actuators	Alerts and priority suggestions
Sensors	Vital signs, symptoms and patient records

Environment Characteristics

- Highly dynamic
- Partially observable
- Non-deterministic
- Multi-agent

Positive Impact

- Helps identify high-risk patients quickly.
- Supports emergency medical staff.
- Can improve prioritization.

Negative Impact

- Incorrect predictions can cause patients to be incorrectly prioritized.
- Bias in historical data can affect results.

Ethical Issue

The system must be tested carefully to avoid unfair treatment of different patient groups.

---

15) Learning Better Medical Diagnosis
Agent
Learning Agent
Environment
Remote/Telemedicine Environment
The AI learns from medical information received remotely.
Examples:

- Patient symptoms
- Uploaded medical reports
- Images
- Previous medical history
- Vital-sign information

PEAS

PEAS	Description
Performance	Accurate and useful remote diagnostic support
Environment	Telemedicine
Actuators	Alerts, reports and recommendations
Sensors	Patient-provided information, reports and images

Environment Characteristics

- Partially observable
- Dynamic
- Non-deterministic
- Multi-agent
- Communication-dependent

Positive Impact

- Helps people in remote locations.
- Provides faster access to medical information.
- Supports doctors during remote consultations.

Negative Impact

- Remote data may be incomplete.
- Poor internet or image quality can affect diagnosis.
- Patients may misunderstand AI recommendations.

Ethical Issue
AI should clearly communicate its limitations and should not present an uncertain prediction as a confirmed diagnosis.

---

4. Common Positive Societal Impact of AI-Based Medical Diagnosis

1. Faster Diagnosis
AI can analyze large amounts of medical information quickly.
2. Better Healthcare Accessibility
AI-based systems can support healthcare services in remote areas.
3. Reduced Workload
AI can help doctors and medical staff analyze large amounts of data.
4. Early Detection
AI can help identify possible disease patterns at an early stage.
5. Better Decision Support
AI can provide additional information to healthcare professionals.
6. Continuous Monitoring
AI can monitor patient information continuously when appropriate.

---

5. Negative Societal Impact

1. Privacy
Medical information is highly sensitive.
Patient data may include:

- Name
- Medical history
- Test results
- Images
- Medications
- Diagnosis

Therefore, strong data protection is required.
2. AI Dependence
Doctors and patients may become too dependent on AI recommendations.
3. Incorrect Diagnosis
An incorrect AI prediction can potentially affect patient treatment.
4. Job Concerns
AI may change the roles of some healthcare workers, especially in highly repetitive diagnostic tasks.
5. Digital Divide
People without access to modern technology may receive fewer benefits.
6. Data Security
Medical databases can become targets for cyberattacks.

---

6. Ethical Concerns

6.1 Privacy
Patient information must be protected from unauthorized access.
6.2 Confidentiality
Medical information should only be available to authorized people.
6.3 Bias
AI trained on incomplete data may perform differently for different groups.
6.4 Discrimination
AI should not provide lower-quality healthcare recommendations to particular groups.
6.5 Accountability
There should be clear responsibility when an AI-supported decision causes harm.
6.6 Transparency
Doctors should understand the purpose and limitations of AI recommendations.
6.7 Human Judgment
AI should support medical professionals rather than blindly replacing professional judgment.
6.8 Cybersecurity
Medical AI systems must be protected against unauthorized access and manipulation.
6.9 Patient Consent
Patients should be appropriately informed when AI is being used in their healthcare.

---

7. Bias and Discrimination in Medical AI

Bias is an important issue in AI-based medical diagnosis.
Suppose an AI model is trained mainly using data from one population.
It may not perform equally well for:

- Different age groups
- Different genders
- Different geographic populations
- Different skin tones
- Different medical conditions

Example
An AI system designed to detect skin disease should be tested using a diverse range of skin tones.
Ways to Reduce Bias

- Use diverse training datasets.
- Test AI on different patient groups.
- Regularly evaluate performance.
- Use qualified medical professionals for validation.
- Monitor performance after deployment.

8. Privacy and Surveillance

Medical AI can process large amounts of personal information.
Potentially sensitive information includes:

- Medical history
- Diagnosis
- Test results
- Medical images
- Medication information
- Patient identity

If this information is improperly collected or shared, it can create serious privacy problems.
Therefore:

- Patient information should be protected.
- Access should be restricted.
- Data should be used only for legitimate purposes.
- Sensitive information should be securely stored.

---

9. Legal Issues

AI-based medical diagnosis creates several legal questions.
1. Responsibility
If AI provides an incorrect recommendation:
Who is responsible?
Possible parties could include:

- Doctor
- Hospital
- AI developer
- Medical-device/software company

2. Patient Data Protection
Healthcare organizations must follow applicable data-protection and healthcare regulations.
3. Medical Approval
AI systems used for medical purposes may require appropriate regulatory approval depending on their intended use and jurisdiction.
4. Transparency
Healthcare professionals should understand the capabilities and limitations of the AI system.

---

10. Final Conclusion

Artificial Intelligence and Machine Learning are becoming important tools in modern healthcare.
This report analyzed **AI-based medical diagnosis** using five AI agents and five different environments, producing **15 different agent–environment cases**.
The analysis shows that different AI agents can support different medical tasks. Simple reflex agents can provide immediate alerts, state-based agents can use patient history, goal-based agents can work toward diagnostic objectives, utility-based agents can compare different options, and learning agents can improve their predictions using medical data.
AI-based medical diagnosis can provide many benefits such as:

- Faster diagnosis
- Early detection
- Better healthcare accessibility
- Reduced workload
- Continuous monitoring
- Decision support for doctors

However, it also creates important challenges such as:

- Patient privacy
- Data security
- AI bias
- Discrimination
- Incorrect diagnosis
- Accountability
- Legal responsibility
- Overdependence on AI

The most important principle is that **AI should support healthcare professionals rather than blindly replace human judgment**.